

CIS255 Final Project Narrative Overview

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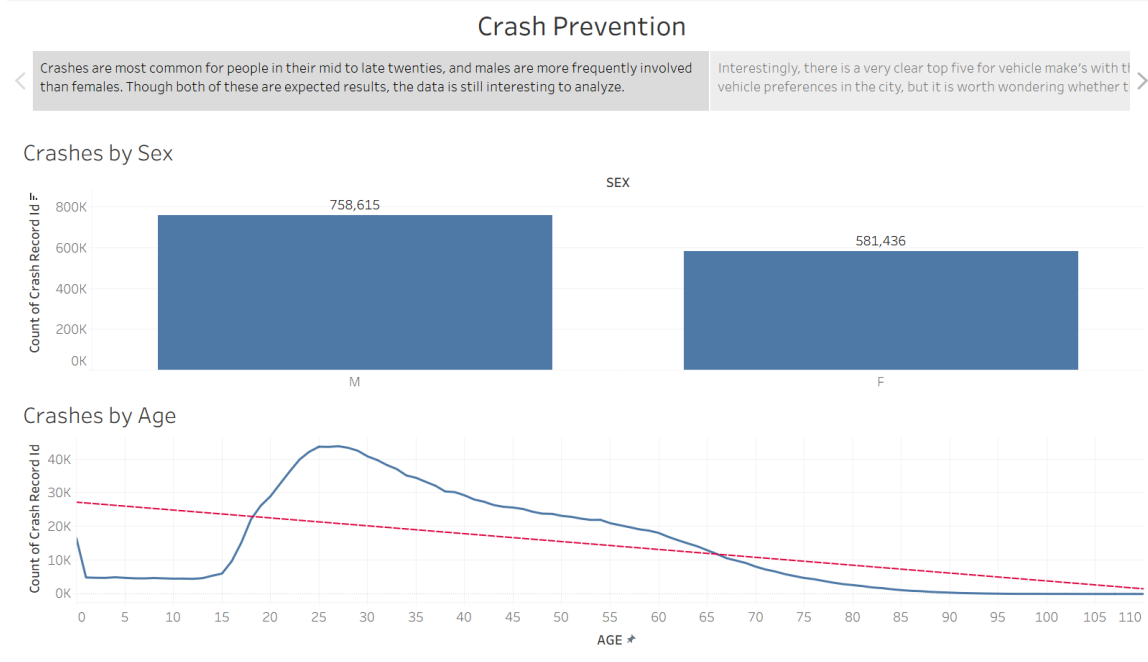
Overview:

Reducing Traffic Crashes in Chicago: A Data-Driven Analysis of Urban Road Safety

For our final data visualization project, We explored traffic crash data from the City of Chicago's open data portal, focusing on crash patterns across demographics, vehicle types, locations, time periods, road conditions, and contributing causes. Our goal was to identify the key factors driving crashes in Chicago and provide actionable, data backed insights that city planners, traffic engineers, and policymakers could use to make Chicago's streets safer for drivers, cyclists, and pedestrians alike.

The dataset, sourced from the Chicago Data Portal, captures thousands of reported crashes across the city, offering a rich foundation for uncovering trends. It is split into three separate components which we joined together, one with general information about each unique crash, one with data about the vehicles involved in each crash, and the other with data about the people involved. Through this analysis, we explored questions such as how crash frequency shifts by time of the day, season, and year; which vehicle makes appear most often in crash reports; what the most dangerous locations in the city are and why intersections account for such a disproportionate share of incidents; and how weather and road conditions amplify crash risk. We also examined the demographic breakdown of those involved and investigated the most commonly cited causes of crashes and injuries.

1. How does crash data vary by the demographics of the people involved?

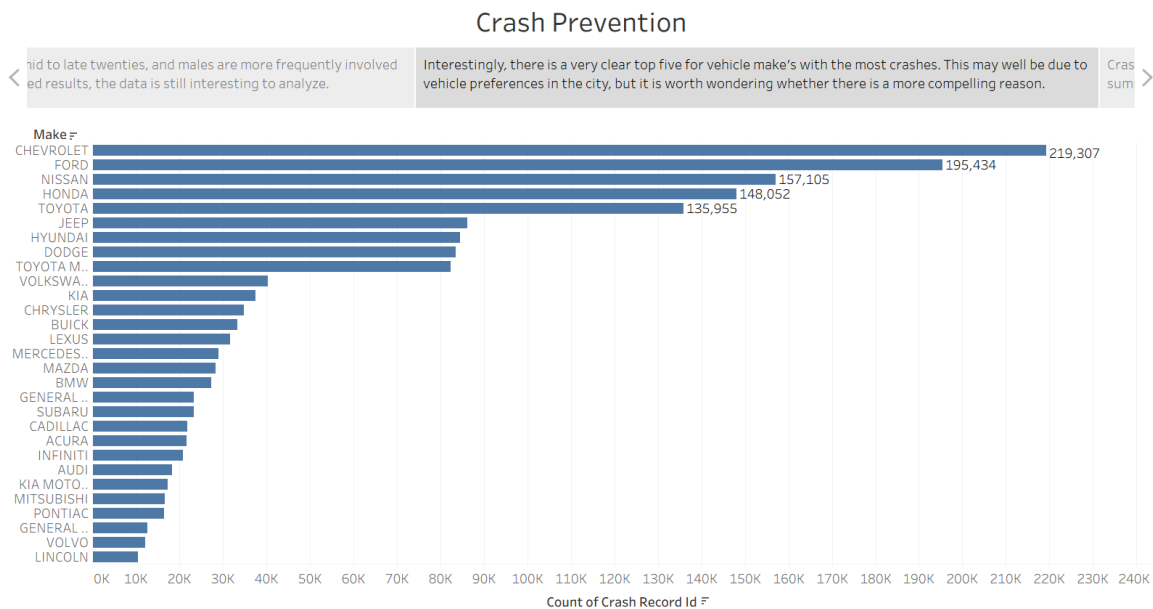


Crash Demographics: Age and Sex of Those Involved

The first visualization examines how crash involvement in Chicago breaks down across two key demographic variables: sex and age. The bar chart makes the gender gap immediately visible, with males involved 758,615 crashes compared to 581,436 for females. Meanwhile, the age line chart tracks crash frequency across every age group, revealing a dramatic spike beginning at age 16 when most drivers first get their licenses, climbing to a sharp peak around age 27, before gradually tapering off through middle age and dropping significantly past 65.

What makes this visualization particularly useful is that it moves the conversation beyond raw crash totals and into the question of who is actually behind the wheel when these incidents happen. The combination of both charts on one dashboard allows viewers to see that the problem is not evenly distributed across the population. Young male drivers represent a concentrated area of risk, and that concentration has real implications for where prevention efforts should be directed. Targeted interventions like stronger graduated licensing laws, mandatory defensive driving courses for new drivers, and public awareness campaigns aimed at the 16 to 30 age range are not just logical conclusions from this data, they are the kinds of policy decisions that this visualization was designed to support. Understanding who is crashing is the first step toward understanding how to stop it.

2. How does the number of crashes change by vehicle make?

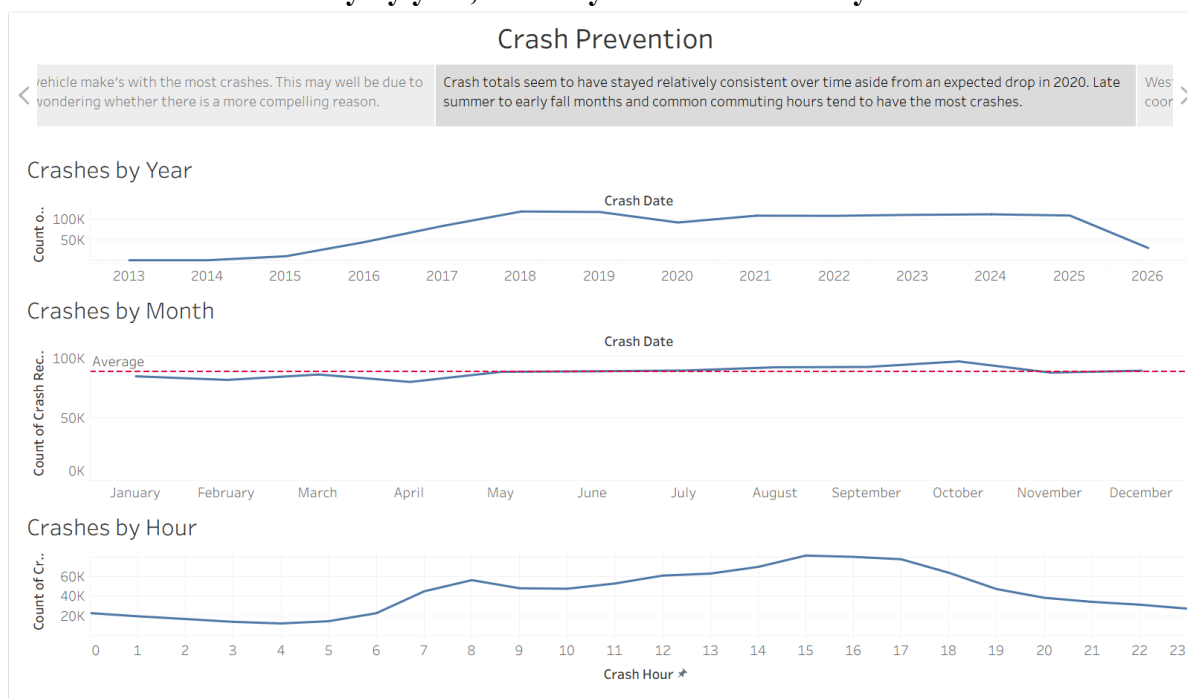


Crashes by Vehicle Make: The Dominance of Domestic Brands

The horizontal bar chart ranks every major vehicle make in Chicago's crash dataset by total crash involvement, and the results show a clear hierarchy at the top. Chevrolet leads all manufacturers with 219,307 crashes, followed by Ford at 195,434, Nissan at 157,105, Honda at 148,052, and Toyota at 135,955. These five brands alone account for a massive share of all recorded crashes in the city, while every other make on the list falls well below the 90,000 mark, creating a steep drop off that is immediately visible in the chart.

The concentration of crashes among these five makes is significant because they represent the most widely owned vehicles in Chicago, meaning the city's road safety efforts need to be designed with these drivers specifically in mind. Whether that means prioritizing traffic enforcement in areas where these vehicles are most common, working with dealerships to promote safe driving resources at the point of purchase, or using registration data to better target awareness campaigns, this chart makes clear that a small group of vehicle owners is responsible for a disproportionate share of incidents on the road. Understanding which cars are most frequently involved in crashes is not just a statistical curiosity, it is a practical tool for deciding where prevention resources will have the greatest impact.

3. How does crash data vary by year, time of year and time of day?

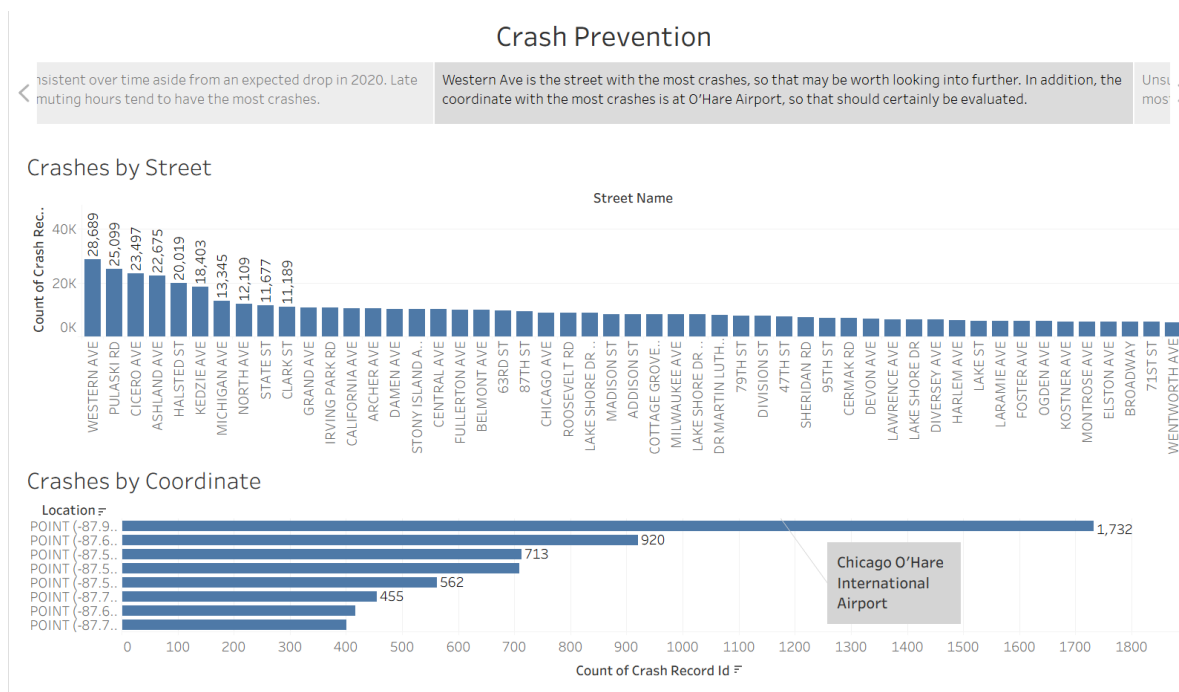


Crash Trends Over Time: Year, Month, and Hour

The three-part visualization breaks down Chicago crash data across different time scales, and each chart tells its own piece of the story. Looking at crashes by year, totals climbed steadily from 2013 and peaked in 2018 at 118,952 crashes before leveling off, with a notable dip in 2020 that clearly reflects reduced traffic during the COVID-19 pandemic. The monthly breakdown shows that crashes are relatively consistent throughout the year but rise noticeably in the fall, with October standing out as the deadliest monthly at 95,745 crashes. The hourly chart is perhaps the most telling of the three, showing crashes hitting their lowest point in the early morning hours before climbing sharply through the workday and peaking at 3pm with 81,325 incidents.

These patterns carry real meanings for how the city should approach crash prevention. The October spike likely reflects a combination of factors including shorter daylight hours, back to school traffic, and shifting weather conditions, all converging in a way that puts more people at risk. The 3pm peak is equally telling because it aligns directly with the end of the school day and the start of the evening commute, two of the highest traffic moments in any city's daily rhythm. Rather than spreading safety resources thin across all hours and months equally, this visualization makes a strong argument for concentrating enforcement, signage improvements, and public awareness efforts during the afternoon commute window and throughout the fall season, when the data consistently shows Chicago's roads are at their most dangerous.

4. What are the most common locations for crashes to occur?

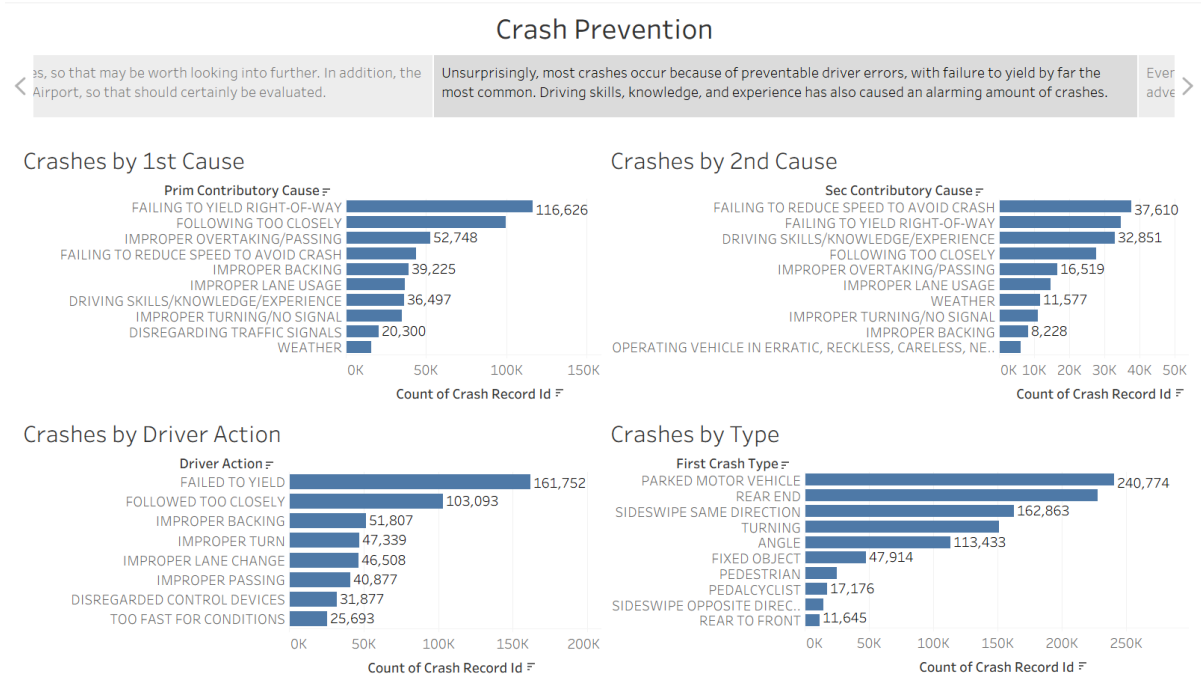


Crash Hotspots: Streets and Coordinates Across Chicago

The location visualization combines two views of the same problem, ranking crashes both by street name and by geographic coordinate to pinpoint where Chicago's most dangerous areas are concentrated. Western Avenue tops the street chart with 28,689 crashes, followed closely by Pulaski Road at 25,099, Ashland Avenue at 23,497, Halsted Street at 22,675, and Kedzie Avenue at 20,019. These are all major north south arterial roads that run the full length of the city and carry enormous daily traffic volume. The coordinate chart adds another dimension entirely, identifying a single point near O'Hare International Airport as the location with a staggering 1,732 crashes, far outpacing every other coordinate in the dataset.

The fact that Chicago's most crash prone locations are its longest and busiest corridors is not a coincidence. Streets like Western and Pulaski stretch across dozens of neighborhoods, passing through countless intersections, bus stops, and commercial zones that create countless opportunities for conflicts between vehicles, cyclists, and pedestrians. These are roads that were built for throughput, not necessarily for safety, and the data reflects that tension directly. The O'Hare coordinate finding is particularly worth investigating further, as that level of concentration at a single point suggests a specific structural or traffic flow issue around the airport that goes beyond general volume. Together these two charts give city planners a clear and prioritized list of where infrastructure investment, redesigned intersections, and targeted enforcement would deliver the most immediate reduction in crashes.

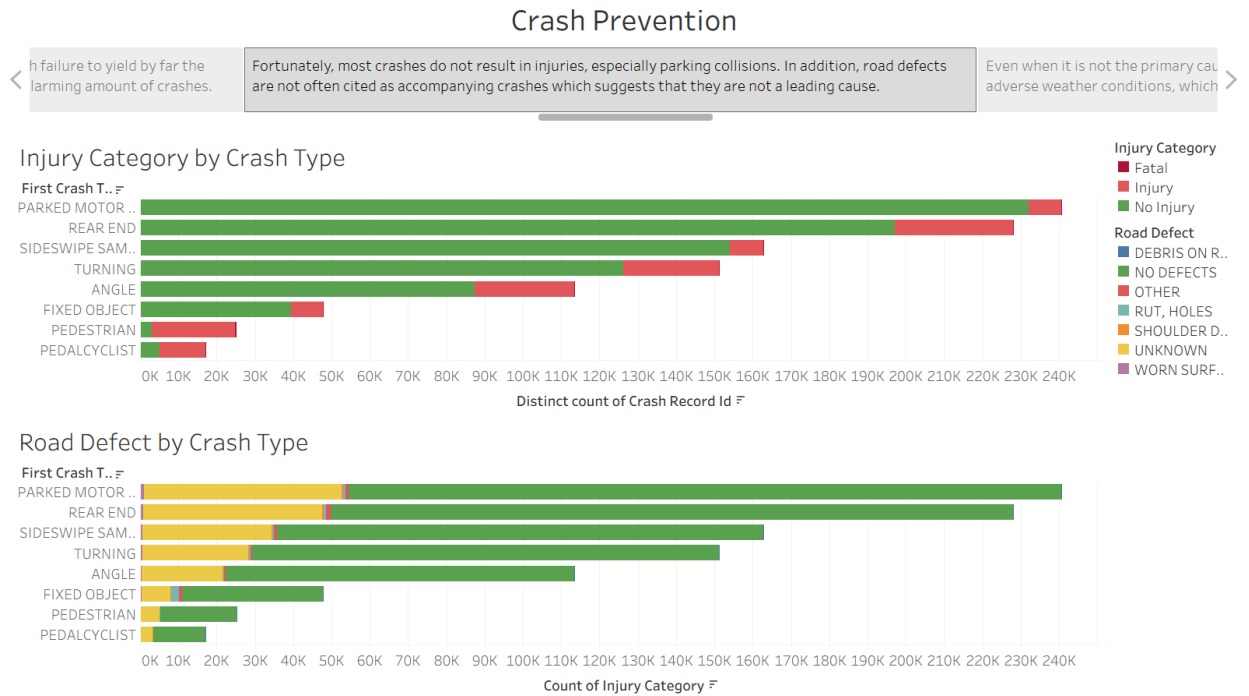
5. What are the most common causes of crashes?



Crash Causes: Preventable Driver Errors Dominate

The data indicates that most crashes occur due to preventable driver errors. The most frequent primary cause is a failure to yield the right-of-way, which accounted for 116,626 incidents. When looking specifically at driver actions, “Failed to Yield” and “Following Too Closely” are the dominant behaviors leading to accidents, resulting in 161,752 and 103,093 crashes respectively. Furthermore, lack of driving skills, knowledge, and experience is cited as an alarming factor. These behaviors often manifest in the most common crash types: hitting a parked motor vehicle (240,774 incidents) or rear-end collisions (162,863 incidents). There seems to be a need for improved driver education or stricter requirements to earn a license, as many of these crashes could have been avoided.

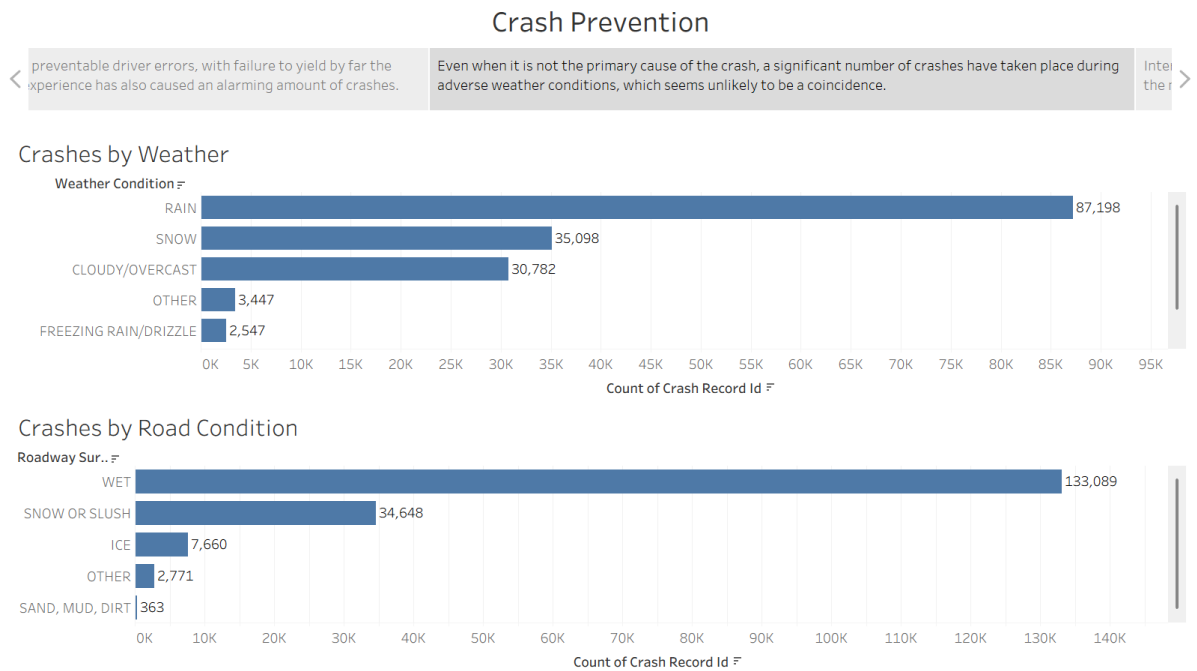
6. How does the outcome of the injury get affected by road defects?



Road Defect and Severity Analysis: A Potential Problem Area Cleared

The data shows that the vast majority of crashes occur under normal roadway conditions with no defects, and most of these incidents result in no injury. Road defects such as potholes or worn surfaces appear in a much smaller portion of crashes and do not show a clear increase in severe injury outcomes. Overall, this suggests that road defects have limited influence on injury severity, while other factors like driver behavior and crash type are likely more important. While proper road maintenance is still important, clearing this potential problem area will allow the city to focus anti-crash initiatives elsewhere.

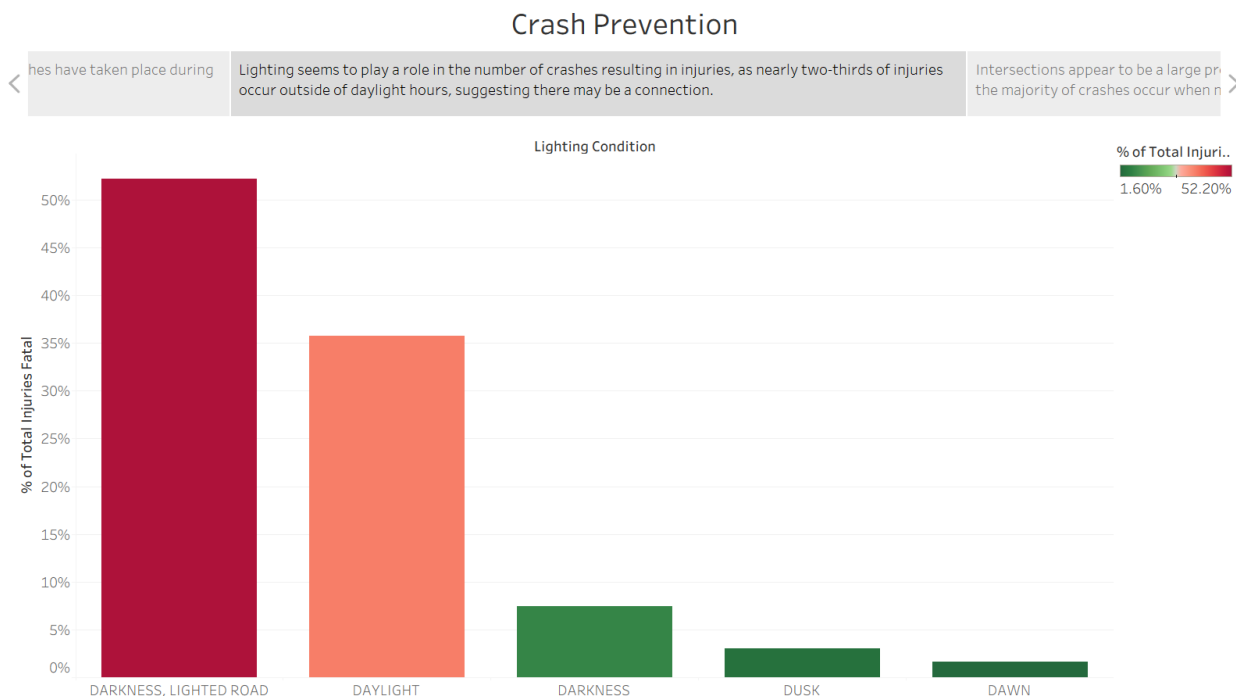
7. How do weather and road conditions impact the number of crashes?



The Importance of Conditions: Weather's Noticeable Contribution to Crashes

Environmental factors play a significant role in road safety, even when they aren't cited as the primary cause. Rain is the most dangerous weather condition captured in the data, linked to 87,198 crashes. Snow and cloudy/overcast conditions also show a notable impact, together contributing to over 65,000 incidents. The state of the roadway itself is equally telling; "wet" road surfaces were a factor in 133,089 crashes, while "snow or slush" accounted for 34,648. This suggests that adverse weather conditions and the resulting road hazards create a high-risk environment for Chicago drivers, and they should be more aware of the dangers of driving in less-than-ideal conditions.

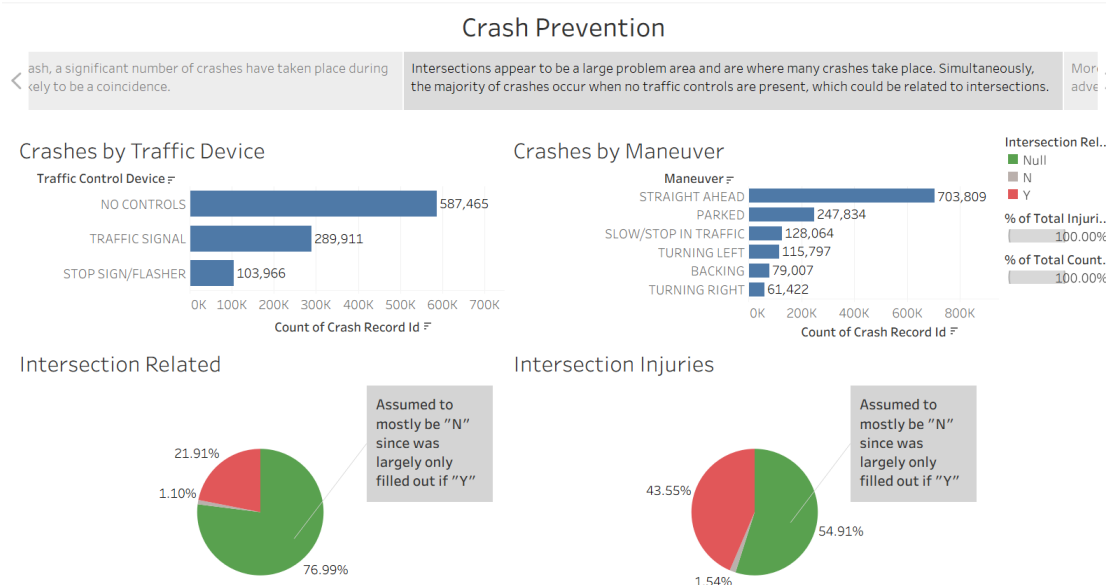
8. Why do crashes occurring in poor lighting conditions tend to result in higher injury rates compared to those in daylight?



Driving at Night: The Most Prominent Time for Fatal Crashes

The data reveals that crashes occurring in "Darkness, Lighted" conditions account for over 50% of total fatal injuries, which is significantly higher than the 35% occurring during broad daylight. This suggests that artificial lighting may create a false sense of security or fail to illuminate hazards as effectively as natural light, leading to higher-impact collisions. Interestingly, "Dawn" and "Dusk" show much lower percentages, likely because the total volume of traffic is lower, but the intensity of the "Darkness" bars points toward visibility being the primary factor in fatality rates. The data clearly shows that reduced visibility from artificial or absent lighting is a primary contributor to higher crash fatality rates in Chicago. As a result, nighttime lighting should be improved when possible throughout the city, and caution should be emphasized when driving at night.

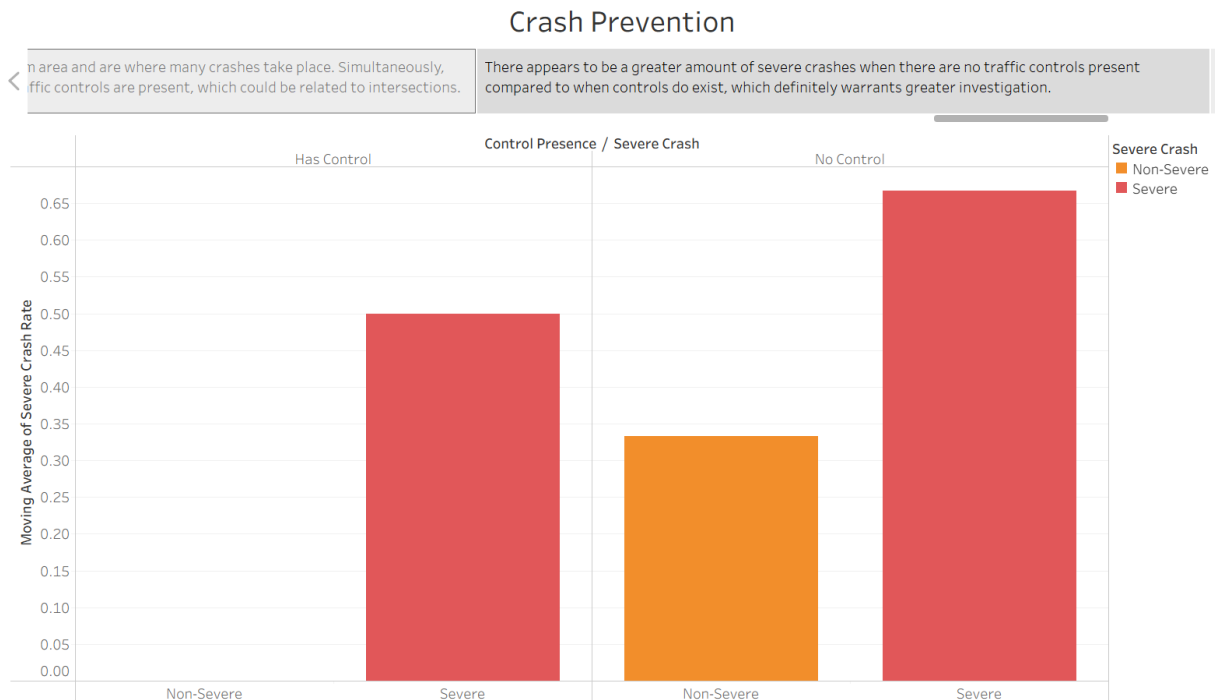
9. Why do such a large percentage of crashes occur at intersections?



Intersections: A Clear Problem Area

Intersections represent a major problem area for urban safety, accounting for a disproportionate share of severe outcomes. While intersection-related incidents make up 21.91% of total crashes (229,062), they are responsible for a staggering 43.55% of all injuries (90,189). Interestingly, the majority of crashes—587,465—occur where there are "No Controls" present, which suggests a possible correlation between unregulated areas and increased risk. Common maneuvers involved in these incidents include going straight ahead, as well as left and right turns, which are typical movements that lead to conflict at intersection points. The city should certainly look into adding more traffic controls at problematic intersections, as there is clearly a need to do so.

10. What is the relationship between the presence of traffic control devices and the rate of severe traffic crashes?



Traffic Controls and Severe Crashes: A Clear Negative Relationship

Locations without traffic control devices exhibit a moving average severe crash rate of approximately 0.67, which is notably higher than the 0.50 rate seen in controlled areas. The visualization demonstrates a clear upward shift in crash intensity when drivers are not guided by signals, signs, or markings. By comparing these two categories, it becomes evident that the risk of a high-impact incident increases as oversight decreases. The data shows that the absence of traffic controls is associated with a higher likelihood of crashes being severe, clearly indicating a need for increased traffic controls in the city.

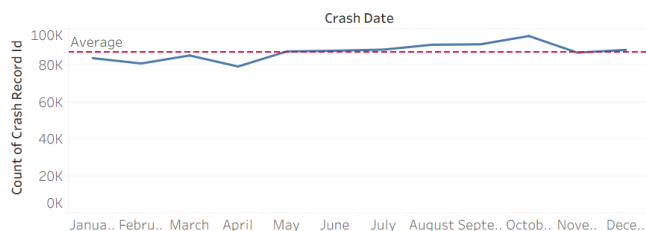
11. How can the number of crashes in the city be reduced to make it safer?

Crash Prevention

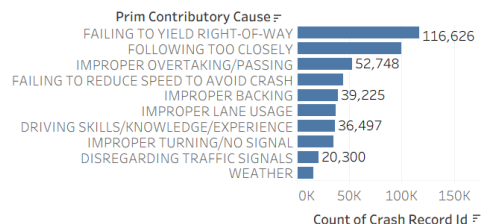
in area and are where many crashes take place. Simultaneously, traffic controls are present, which could be related to intersections.

More police during the summer, more traffic controls (especially at intersections), greater caution during adverse weather, improved driver education, and increased airport controls could help lessen crashes.

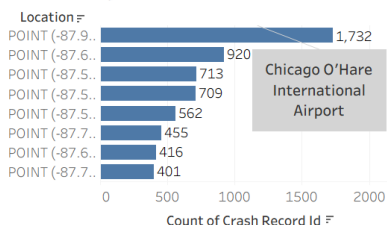
Crashes by Month



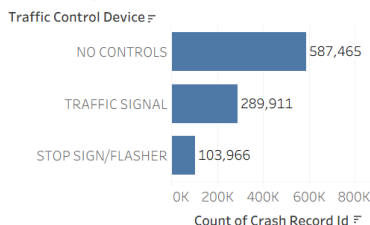
Crashes by 1st Cause



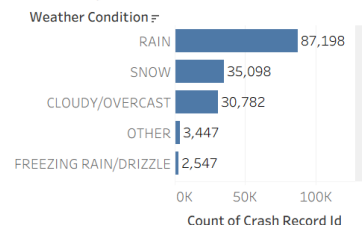
Crashes by Coordinate



Crashes by Traffic Device



Crashes by Weather



Final Analysis: Recommendations to the City to Cut Down on Crashes

To improve city safety, the data supports a targeted approach rather than spreading resources thin. Key recommendations include increasing police presence during the summer months and implementing more traffic controls, specifically at dangerous intersections. Because the data shows a sharp peak in crashes at 3pm, aligning with the end of the school day and the start of the commute, enforcement and public awareness efforts should be concentrated during these afternoon hours. Specific initiatives should be focused on the dangers of driving during adverse conditions as well, as weather and lighting seem to play a role in crashes. Additionally, improving driver education to address skills and knowledge gaps, alongside structural traffic flow evaluations at major hotspots like O'Hare Airport, could provide immediate reductions in the city's crash rate.

Technical Approach

The crash datasets we worked with contain data spanning from 2013-2026 up to the date it was downloaded, with most crashes recorded taking place post-2015. We joined the three datasets on the “Crash Record ID” field, as they are all related and come from the same source, they just contain different fields of data for each unique crash. We used multiple different types of visualizations within Tableau as well as its calculation features to analyze the data and derive greater insights as to what is causing many crashes to take place. We chose multiple colors, filtering, labeling, and average lines to clearly communicate the message of each visualization and make them easy to understand.

Impact and Insights

The purpose of this data story is to not only make Chicago drivers aware of common causes of crashes so that they can be safer on the road but also to give the city of Chicago’s legislature meaningful, data-driven recommendations for how to potentially lessen the number of crashes on city streets. While it is impossible to fully eliminate traffic crashes, any initiative that helps to cut down on the amount of them should be welcomed so that people are safer while in the city. The visualizations reveal that the time of year, weather and lighting conditions, specific location, lack of traffic controls, and poor driver education are all related to higher crash totals, so the city should first look into initiatives that address these areas and analyze the results to determine success rate.

References

[Traffic Crashes - Crashes | City of Chicago | Data Portal](#)

[Traffic Crashes - Vehicles | City of Chicago | Data Portal](#)

[Traffic Crashes - People | City of Chicago | Data Portal](#)